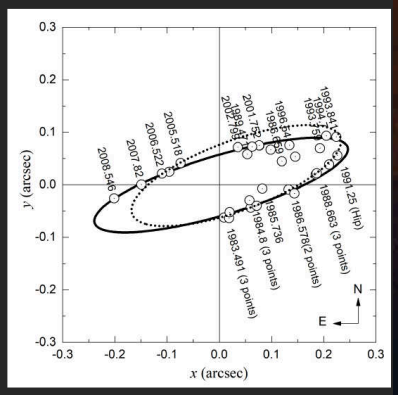
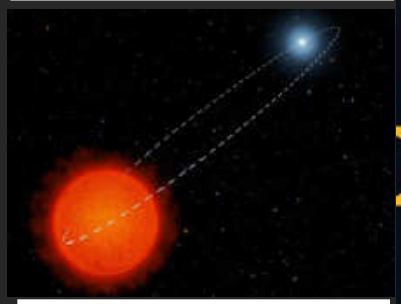
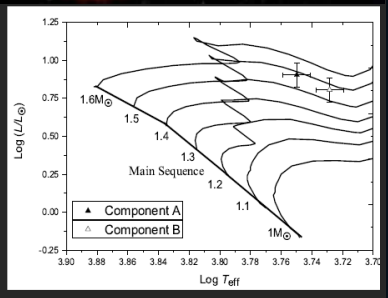
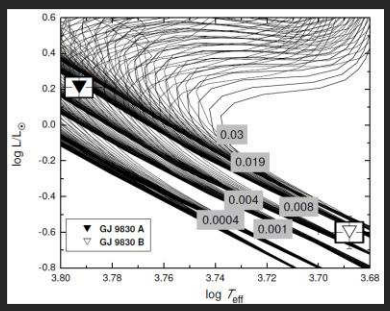










SHARJAH ACADEMY FOR ASTRONOMY, SPACE SCIENCES AND TECHNOLOGY

أكاديمية الشارقة لعلوم وتكنولوجيا الفضاء والفلك

Invitation

SAASST and The Regional Center for Space Science and Technology Education for Western Asia-Jordan/ UN (RCSSTE) in Cooperation with Al al-Bayt University (AABU), and The Arab Union for Astronomy and Space Sciences (AUASS) are hosting,

Online workshop

Binary and Multiple Stars:
From Observations to Estimations

Saturday and Sunday 18th and 19th July 2020

11:00am Abu Dhabi time

ARABIC

Under the patronage of His Excellency **Prof. Hamid Al-Naimiy**
(Chancellor of the University of Sharjah, General Director of SAASST and President of AUASS)

Cooperation with

For More information

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UNIVERSITY OF SHARJAH

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www.SAASST.ae

Day 1: Saturday 18th July 2020

To Login: https://zoom.us/my/saasst

Zoom Login and Registration: to start at 10:30			
Session No.1: Opening Session- Moderator: Prof. Mashhoor Al-Wardat			
Time	Talk	Speaker	Affiliation
11:00-11:10	Welcome speech	Prof. Hamid M. Al-Naimiy	Chancellor: UoS General Director: SAASST President: AUASS
11:10-11:20	RCSSTE speech	Dr. Awni M. Kasawneh	Director: RCSSTE Secretary-General: AUASS
11:20-11:30	AABU speech	Prof. Eqab Rabee	Vice Chancellor for Academic Affairs: AABU
11:30-11:45	Gaia Team	Prof. Martin Barstow	**Professor of Astrophysics:** School of Physics and Astronomy, University of Leicester. **Gaia Team** **Chair**, UK Space Agency Science Programme Advisory Committee. **Chair**, Space Telescope Institute Council **President**, Royal Astronomical Society (2014-2016)
Session No.2: Prelude Moderator: Dr. Alaa Azzam			
Time	Title of the talk	Speaker	Affiliation
11:45-12:00	From Observations to Estimations	Prof. Mashhoor A. Al-Wardat	**Professor of Astrophysics**, Dep. of Applied Physics and Astronomy- UoS. **Academic Coordinator**: SAASST
12:00-12:20	A window to International Cooperation in Binary Stars	Dr. Ali A. Taani	**Associate Professor of Astrophysics,** Dep. of Physics Balqa Applied University - Jordan
12:20-12:45	Classification of Binary Stars	Prof. Hatem S. Widyan	**Professor of Astroparticles**, Dep. of Physics, Al al-Bayt University
Break (15 minutes)			

Session No.3: X-Ray Binaries and Compact Objects Moderator: Dr. Ali Taani			
Time	Title of the talk	Speaker	Affiliation
13:00-13:30	Spectral Behavior of Some Black Hole Binary Systems	Dr. Hanna A. Sabat	**Assistant Professor of Astrophysics,** RCSSTE
13:30-14:00	High Mass X-ray Binaries: Vela X-1	Dr. Antonios Manousakis	**Assistant Professor of Astrophysics**, Dep. of Applied Physics and Astronomy- UoS. **Researcher**: SAASST
Session No. 4: Orbital Analysis – Moderator: Dr. Antonios Manousakis			
Time	Title of the talk	Speaker	Affiliation
14:00-14:30	Two-Body Problem (Theoretical background)	Prof. Hatem S. Widyan	**Professor of Astroparticles**, Dep. of Physics, Al al-Bayt University
14:30-14:45	Orbital Solution & Tokovinin's Dynamical Method [ORBITX code]	Mr. Abdullah Hussain	**M.Sc. St.** Dep. of Physics, Al al-Bayt University
14:45-15:00	Astrophysical data archives of binary Stars: The 4th and 6th Catalogues (Preparing input data)	Mr. Yamam Tawalbeh	**M.Sc. in Physics**: Ministry of Education, Kuwait
15:00-16:00	Solving Orbits (Each participant should choose a binary star and solve its orbit) – Coordinator: Prof. Mashhoor Al-Wardat		
End of the First Day			

Day 2: Sunday 21st July 2019

Session No.5: Spectroscopic Analysis - Moderator: Prof. Husam El-Nasser			
Time	Title of the talk	Speaker	Affiliation
11:00-11:30	Al-Wardat's Method for Analyzing Close Visual Binary and Multiple Stars (CVBMSs): The story	Prof. Mashhoor A. Al-Wardat	**Professor of Astrophysics**, Dep. of Applied Physics and Astronomy- UoS. **Academic Coordinator**: SAASST
11:30-12:00	Ability of Al-Wardat's Method in Investigating Multiplicity of Stars	Ms. Zahra Yousef	**Ph.D. Student**: UKM Malaysia
12:00-12:15	Hipparcos Mission	Ms. Mai Al-Mashaqbeh	**M.Sc. St.** Dep. of Physics, Al al-Bayt University
12:15-12:30	Gaia Mission	Ms. Diala M. Taninneah	**M.Sc. St.** Dep. of Physics, Al al-Bayt University
12:30-12:45	Sixth Catalog of Orbits of Visual Binary Stars	Bayan Hammad	**M.Sc. St.** Dep. of Physics, Al al-Bayt University
12:45-13:00	Future Space Telescopes	Ms. Fatima Alkhateri	**SAASST**
13:00-13:30	Sharjah Observatory and its facilities	Mr. Moh'd Talafha	**M.Sc. in Physics** Researcher: SAASST
13:30-13:45	Observing Binary Stars Using Amateur Telescopes	Mr. Mohammad Aljady	**Amateur Astronomer:** Kuwait
Brake: 15 minutes			

Session No.6: Observations – Moderator: Dr. Ahmad Abu-Shattal			
Time	Title of the talk	Speaker	Affiliation
14:00-14:30	Primary Calculations and Input Parameters for Al-Wardat's Method	Ms. Enas Abu Alrob	**M.Sc. in Physics**
14:30-16:45	Applying Al-Wardat's Method for Analyzing Selected Close Visual Binary and Multiple Stars (CVBMSs)- **Coordinator: Mr. Abdallah Hussein** (Each participant should choose a binary star and solve its orbit) - Installing software - Collecting data - Calculating input parameters - Applying the method - Plotting results		
Brake: 15 Minutes			
Session No.7: Workshop's Guests - Moderator: Prof. Mashhoor Al-Wardat			
Time	Title of the talk	Speaker	Affiliation
17:00-18:00	Data reduction with PyRAF	Dr. Moh'd Mardini	**Ph.D. in Astrophysics**
18:00-18:30	Orbit Calculation	Prof. Andrei Tokovinin	**Prof. of Astrophysics (DrSci)** Cerro Tololo Inter-American Observatory CTIO/AURA, Chile
18:30-19:00	Open Discussion and Closing Remarks	Prof. Mashhoor A. Al-Wardat	**Professor of Astrophysics**, Dep. of Applied Physics and Astronomy- UoS. **Academic Coordinator**: SAASST
End of the Workshop			